MILITARY SPECIFICATION SHEET

VALVE, STARTER CONTROL, AIRCRAFT ENGINE, FOR F404 ENGINE/A6, F-18 AIRCRAFT

This specification is approved for use by the Naval Air Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

The complete requirements for acquiring the starter control valve described herein shall consist of this specification and the latest issue of Specification MIL-V-81995.

REQUIREMENTS:

ENVELOPE: See Figures 1, 2, and 3.

Weight: 7.5 lb maximum including Frequency Sensing Relay (FSR) and Pneumatic Control Unit (PCU).

ATSCV - 5.25 lbs max.
FSR - 1.0 lbs max.
PCU - 1.25 lbs max.

Aircraft Model: A6, F-18.

Engine Model: F404.

Inlet Air Conditions:

Rated Pressure: 65 psia.

Rated Temperature: 455°F.

Maximum Continuous Temperature: 455°F.

Intermittent Temperature: 1160°F (1 minute max.)

Engine Interbreed Pressure:

Normal inlet pressure - 105 psia
Maximum inlet pressure - 175 psia

Minimum Pressure: 15 psig.

Proof Pressure: 205 psia.

Burst Pressure: 395 psia.

FSC 2995

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Regulating Pressure: 48 ±3 psig at Inlet pressures of 65 psia to 105 psia and 48 ±4 psig up to 175 psia at airflows from 50 lb/rein to 200 lb/rein.

Flow. The flow rate for all test conditions shall be limited by a fixed orifice or nozzle which will produce an airflow of 150 ±3 lbs/min. at orifice and nozzle inlet conditions of 48 ±3 psig and 455 ±lO°F.

Rate of Pressure Rise. The initial pressure rise shall be limited to a maximum of 22 psi/see during the first 0.5 second after the first indication of pressure rise and thereafter the pressure rise shall be smooth and at a positive rate.

Opening and Closing Times: Open 5 seconds maximum; close 1.5 seconds maximum and 0.8 second minimum, at 37 psig inlet pressure.

Pressure Drop. Pressure drop shall not exceed 1.25 psid at a flow of 150 lbs/min.

Leakage. Internal and external leakage combined shall not exceed 1.36 lb/rein when the valve is closed. External and bleed leakage combined, when the valve is regulating, shall be 1.81 lbs/min maximum.

Altitude:

a. Maximum operating: 30,000 ft.

b. Maximum non-operating: 55,000 ft.

Ambient Temperature Range:

a. Operating: -65 to +250°F.

b. Non-operating: -65 to +300°F.

Ambient Temperature Range FSR:

a. Operating ground: -65 to +135°F.
Operating flight 2 min. max: -65 to +160°F.

b. Non-operating: -80 to +203°F.

Flight and Ground Loading:

a. Flight maneuvering: 30g.

b. Shear: 50 lb.

c. Axial: 100 lb.

d. Moment: 300 in-lb.

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Vibration: See Figure 4.

Military Part Number. The military part number of the valve shall be M81995/6.

Position Indicator. The position indicator switch shall operate before 16° open in the opening direction and before 6° open in the closing direction.

External Surface Temperature. The external surface temperature of the ATSCV shall not exceed 400°F during or following any operating condition, whenever the ambient temperature is 350°F or less. Provisions necessary to meet this requirement may be satisfied by use of an external covering or insulating blanket.

Pneumatic Control Unit (PCU). The PCU shall be remotely located from the valve and FSR with interconnecting tubing and electrical wiring to be furnished by the aircraft manufacturer.

Performance: The bleed flow from the PCU versus power input using a nominal 39 ohms coil resistance shall conform to Figure 5.

Life: The life test of the PCU shall consist of a minimum of the following cycling. The current shall be cycled from 0 to 300 mA and back to zero in three minutes. The conditions shall be as shown below:

Con	Ambient Temperature	Inlet Temperature	Inlet Pressure	Cycles
1	-40°F	265°F	37 psig	100
2	90°F	390°F	31 psig	1,400

Frequency Sensing Relay (FSR). The FSR shall be furnished as a remotely mounted unit and shall meet the requirements of MIL-E-5400 where not in conflict with criteria specified elsewhere in this specification. All inputs and outputs shall terminate at a single connector.

FSR/PCU. The FSR shall interface with the PCU and the ATS to provide the following controls to the ATSCV:

a. Close the ATSCV on increasing speed at 64 ±1 percent (16,140 ±250 rpm) of ATS output speed.

b. Maintain 59 ±1 percent ATS speed during system checkout when the AMAD is decoupled from the main engine.

ATS Speed Signal. The electrical input derived from the ATS speed sensor shall result in loss of nominal 28 vdc power at the FSR output when the input frequency reaches 6,354-6,556 Hz (10,926 rpm PTS speed). The maximum amplitude expected is 30 volts peak-to-peak. The FSR shall meet performance and BIT requirements with a maximum external capacitance of 14 nF between each speed sensor line and ground.

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ATS Speed Signal Threshold. The speed signal inputs of the FSR shall be balanced through appropriate switching for left or right activation. The signal threshold (minimum amplitude required for operation) shall be as low as possible, consistent with EMIC and BIT requirements. The threshold shall not exceed 0.55 volts rms for a sine waveform at 1,000 Hz, and it shall not exceed 0.75 volts rms at 6,360 Hz due to input circuit filtering.

Input Impedance - Speed Sensor. The speed signal input impedance, when selected, shall be greater than 1,400 ohms at 6,500 Hz. When not selected, the input impedance shall exceed 10,000 ohms. Measurement shall be made at room temperature.

Zero Speed Input. If a speed signal of not less than 1,000 Hz is not sensed at 12 ±4 seconds, the FSR output shall be disabled.

ATS Speed Output. A buffered, conditioned speed signal shall be provided at the FSR connector.

Zero Speed Shutoff. If the FSR, after energizing the crank switch holding coil, fails to detect a speed signal of greater than 806 Hz, it shall de-energize the crank switch holding coil after 12 ±4 seconds have elapsed.

FSR Output. The output shall be capable of continuously energizing the crank switch holding coil. Rated coil hold-in voltage is 15 to 30 vdc. Minimum coil resistance is 220 ohms at 20°C.

FSR Output Signal to PCU. The PCU output shall control current through the PCU in accordance with the following characteristic equation:

$$I_{PCU} = .4166\,(f - 5{,}650) + .125\,\frac{d}{dt}(f) \quad (mA)$$

where f = speed signal frequency modified by appropriate dynamic response characteristics to provide optimum system response.

Start Signal. Electric power in conformance with MIL-STD-704, Category B, will be supplied from engine start circuitry to the left crank or right crank inputs of the FSR. Application of power shall initiate BIT. If no fault is detected, nominal 28 vdc power shall be connected to the output.

Warm-Up Requirements. No warm-up is allowed.

Ground Maintenance Mode Speed Control:

The ATSCV shall regulate ATS speed during ground maintenance operation to 59 ±1% (100% = 16,810 rpm PTS speed) during all steady state conditions in which the ATSCV is less than full open.

The maximum ATS speed during start-up and step load decreases shall not exceed 64 percent.

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The minimum ATS speed following a step load decrease shall not be less than 49 percent.

Operating Conditions:

Maximum ATS Accessory Load	90HP
Minimum ATS Accessory Load	25HP
Maximum Step Load Change	59HP
System Inertia Reference to ATS	0.04135 (slug-ft)
Duct Volume Upstream of ATSCV	910 IN^3
Duct Volume Downstream of ATSCV	148 IN^3
ATSCV Air Inlet Conditions	Rated
ATS Torque Characteristics	Rated

The system shall function in the aircraft and be compatible with the air supply system to provide speed control while the aircraft is on the ground.

Speed control shall not operate in other than ground maintenance mode.

FSR & PCU Moisture Design and Packaging. The FSR shall be designed to minimize the collection of moisture encountered when the equipment bay doors are open. Use of seals, baffles, and placement of internal components shall be considered to improve moisture protection. Modes of entrance of moisture shall be studied and eliminated as much as practicable. Sufficient area shall be provided for drainage of entrapped moisture, with equipment in the normally installed position.

The FSR will be considered non-repairable and may be hermetically sealed. No adjustments are required.

PCU Control. The FSR shall incorporate a single output providing low side (28 vdc return) control of the selected PCU. The right or left PCU circuit will be selected through external switching provided by the aircraft.

Interchangeability. The ATSCV and its components, FSR and PCU, shall be physically interchangeable or replaceable with any other manufacturer listed on the applicable Qualified Products List (QPL).

Useful Life. The equipment shall have a useful life of not less than 15 years under any combination of operating and storage life without exceeding the operational service life.

Operational Service Life. The equipment shall have an operational service life of not less than 7,800 hours including 6,300 main engine starts and 2,400 ground maintenance operations of 30 minutes duration under any natural combination of environmental conditions specified herein. Service life for the purpose of this equipment shall hereafter be considered in terms of

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equivalent cycles as defined in Life Cycle. Operational service life is defined as the total operating time between the start of operation and wearout. Wearout is defined as the point where overhaul or repair cost exceeds one-half of the original cost of the equipment.

Life Cycle. The equipment shall be capable of meeting the requirements of this specification during and following 7,800 hours of operation.

Storage Life. Storage life capability of the equipment shall be no less than 36 months. Storage life is the period for which the equipment may be kept in storage before installation and still have the specified operational service life after installation without component or part replacement, adjustment, or maintenance action.

Service Condition, Electrical. The equipment shall give specified performance when supplied with electric power having characteristics and limits as defined in MIL-STD-704, Category B. The 50 milliseconds power interruption requirement is waived.

Dielectric Strength. The equipment, excluding the FSR, shall be capable of withstanding, without damage or breakdown, a minimum of 25 applications of the applicable voltages shown below, at commercial frequency, for one minute between mutually insulated parts. Any arc-over (air discharge), flash-over (surface discharge), or breakdown (puncture discharge) at the specified voltages shall constitute failure. Failure may be evidenced by fluctuations in the leakage current, a steady increase in leakage current, or more than twice the normal leakage current at the specified voltage. The normal leakage current shall be specified for each test point for the design finally chosen. The normal leakage current shall be established by test on a minimum of six units.

SYSTEM VOLTAGE	VOLTAGE (Standard Conditions)	VOLTAGE (70,000-Foot Altitude)
28 Vdc	1050	500

Capacitors which are not integral parts of the FSR shall be capable of withstanding for one minute twice the maximum peak voltage encountered during normal operation. Diodes used for coil suppression shall have a Peak Inverse Voltage (PIV) rating of 800 volts minimum.

Bonding. A means of electrically bonding the equipment to the aircraft shall be provided in accordance with MIL-B-5087, Class R. All contact surfaces of the ATSCV case (brackets, connectors, flanges, panels, etc.) shall be treated per MIL-STD-889. The ATSCV shall be protective treated per MIL-STD-889 at the bond interfaces.

Insulation Resistance. The insulation resistance of electrical components, excluding the FSR, shall not be less than 100 megohms when exposed to 500 ±10 volts dc applied between all mutually insulated circuits for two minutes duration for all environmental conditions specified herein.

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BIT General Requirements. When electric power is applied to the left or right crank input of the FSR, a test signal shall be momentarily applied through the speed sensor circuit corresponding to the crank input energized. The test signal shall simulate the speed signal in such a way as to activate the FSR speed logic to effect an overspeed condition. If the FSR responds correctly to the simulated overspeed, input power shall be immediately connected to the output. The output shall remain energized until input power is removed, or an overspeed condition is sensed or the speed signal becomes inadequate.

Test Time. The test shall be completed within 160 milliseconds.

Reliability. The equipment reliability shall be designed-in through the application of the specific requirements defined herein and other associated documents. The equipment, including BIT, shall have a specified Mean-Cycles-Between Failures (MCBF) of 12,500 cycles for the ATSCV, 37,500 cycles for the FSR, 60,000 cycles for the PCU, for a combined ATSCV ship set MCBF of 5,556 cycles (0° as defined in MIL-STD-781). The equipment shall provide this MCBF when operated in any mode or combination of modes and under any natural combination of the loads and environmental conditions specified herein. In meeting this requirement, the scheduled preventive maintenance by Navy for service usage will be permitted. Successful completion of the testing herein shall not relieve the Seller of the responsibility for compliance of production units with the specified reliability and performance requirements during any subsequent testing or service usage within the limitations specified herein.

Maintainability. The equipment shall be designed in accordance with the qualitative and quantitative maintainability requirements stated herein. The equipment shall be capable of being maintained at Organizational, Intermediate, and Depot levels of maintenance as defined in OPNAVINST 4790.2.

Direct Maintenance Manhours/Flight Hour (DMMH/FH). The equipment shall be designed such that the DMMH/FH measured at the end of Full Scale Development (FSD) and subsequently thereafter through the Fleet Supportability Evaluation (FSE) shall not exceed 0.0052 DMMH/FH (Scheduled and Unscheduled Organizational and Intermediate Level of Maintenance).

Mean Time Between Unscheduled Maintenance Actions (MTBUMA). The equipment shall be designed such that the MTBUMA measured at the end of FSD and subsequently through FSE shall be not less than 1,200 flight hours. MTBUMA is defined as the number of unscheduled maintenance actions occurring in a time interval divided into the total aircraft flight hours in the same time interval. Maintenance action consists of all tasks resulting from a malfunction required to return the aircraft and its equipment to operating condition.

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Electromagnetic Interference and Compatibility (EMIC). The equipment shall comply with MIL-STD-461, with the following changes and additions. Requirements CE03, CE04, CS01, CS02, CS06, RE02, RS02, and RS03 shall apply, except the field intensity level requirement of RS03 is changed to:

20 volts/meter from 14 KHz to 200 MHz
65 volts/meter from 200 MHz to 450 MHz
20 volts/meter from 450 MHz to 1.0 GHz
100 volts/meter from 1.0 GHz to 12.0 GHz
2 volts/meter from 12.0 GHz to 40.0 GHz*

*Except that 100 volts/meter at 16.5 GHz and 33.0 GHz

Transient Susceptibility. No degradation of performance shall occur when the system's interconnecting and signal leads are bundled throughout their entire length with a wire carrying a 600 volt peak-to-peak relay switching transient.

Spike Emission. Spikes (transients of duration less than 500 microseconds) generated by the equipment shall not exceed the following values when measured from the base of the transient:

a. dc power lines: ±50 percent or -150 percent of the normal dc line voltage.

b. ac power lines: +50% of the norms rms voltage

c. Other lines: limits shall be specified in the approved EMC qualification test procedure.

Ground Plane Considerations. All interface circuits shall be capable of specified performance when subjected to the following aircraft chassis noise existing between any two WRAs:

a. 3 volts rms from 320 Hz to 500 Hz

b. 1 volt rms from 10 Hz to 100 MHz

c. ±8 volts, 100 usec pulses at a repetition rate between 100 Hz and 200 Hz

Electromagnetic Compatibility (EMC) Engineering Evaluation. An EMC engineering test and evaluation shall be performed and substituted for the MIL-STD-461, Paragraph 4.2 requirements for an interference control plan preparation and submittal.

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Burn In. Each ATSCV component which includes an electronic part shall be subject to the following burn-in procedure, which includes six complete temperature cycles between -67 to +158°F and ten minutes of vibration while stabilized at each temperature extreme. Complete functional testing shall be performed at each temperature extreme. Temperature, vibration, functional testing, and time requirements are as follows:

a. Temperature extremes shall be maintained at -71 to -63°F (-57 to -53°C) on cold cycles and 154 to 162°F (68 to 72°C) on hot cycles.

b. The time required on the cold cycle shall include twenty minutes minimum stabilization after reaching the cold temperature range and before beginning vibration and functional testing. Transition from hot to cold shall not exceed one hour.

c. The time required on the hot cycle shall include one hour and forty minutes minimum stabilization after reaching the hot temperature range and before beginning vibration and functional testing. Transition from cold to hot shall not exceed forty minutes.

d. Units shall be non-operative during transition and stabilization on the cold temperature cycles. They shall be operative during transition and stabilization on the hot temperature cycles, being activated with right and left starts on alternate cycles.

e. Vibration and functional testing shall be concurrent after the stabilization period at each temperature extreme. Vibration requirements are 2.5 ±0.5 G's at a frequency of 35 ±10 Hz for 10 minutes minimum.

Vibration. Components shall be subjected to a vibration test in accordance with MIL-STD-810, Method 514, Equipment Category b.2, Procedure IA. Tests shall be conducted in accordance with Table 514-IIA and curve "A" of Figure 514-2A. Components may be tested in test assemblies or as individual units. During test, the component shall be subjected to its maximum limiting component temperature specified in the model specification.

Individual Test/Acceptance Test:

Additional tests required are:

A. Valve

1. Switch Operation
2. Insulation Resistance
3. Dielectric Strength
4. Proof Pressure
5. Production Duty Cycle per Table I
6. Calibration
7. Examination

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B. PCU

1. Dielectric Strength
2. Insulation Resistance
3. Proof Pressure
4. Leakage
5. Performance
6. Examination

C. FSR

1. Bonding Resistance

2. Burn-In including vibration: before, during, and after the burn-in, the following functional tests should be performed:

   a. Open Sensor & No Signal Detection
   b. BIT Verification
   c. Dynamic Response
   d. PCU Current Gain & Calibration
   e. Cut-off Frequency & Power Interrupt
   f. Min. Signal Required

3. Examination

Preparing Activity
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TABLE I. Production duty cycle.

Cond.	Operation Mode	Valve Amb at Start of Cycle °F	Air Flow Temp. °F	Inlet Press. Psig	Duration of Flow Seconds per Cycle	Cooldown Valve Body Temp at end of Flow Cycle °F Solenoid De-Energized
I	APU Main Engine Start	Prevailing lab temp. conditions	425 ± 10	37 ± 3	36 +5 -0	90 ± 150
II	Repeat I for a total of 25 times.					
III	Motoring	Prevailing lab temp. conditions	425 ± 10	37 ±3	300 +30 -0	90 ± 150
IV	Repeat III for a total of 2 times.					
V	Ground Checkout (PCU Port) *	Prevailing lab temp. conditions	425 ± 10	37 ±3	300 +30 -0 Total	90 ± 150

*With PCU port adjusted to obtain regulation pressure of 20 ± 10 psig, valve "A" shall be cycled ON for 10 ± 5 seconds and OFF for 10 ± 5 seconds continuously.

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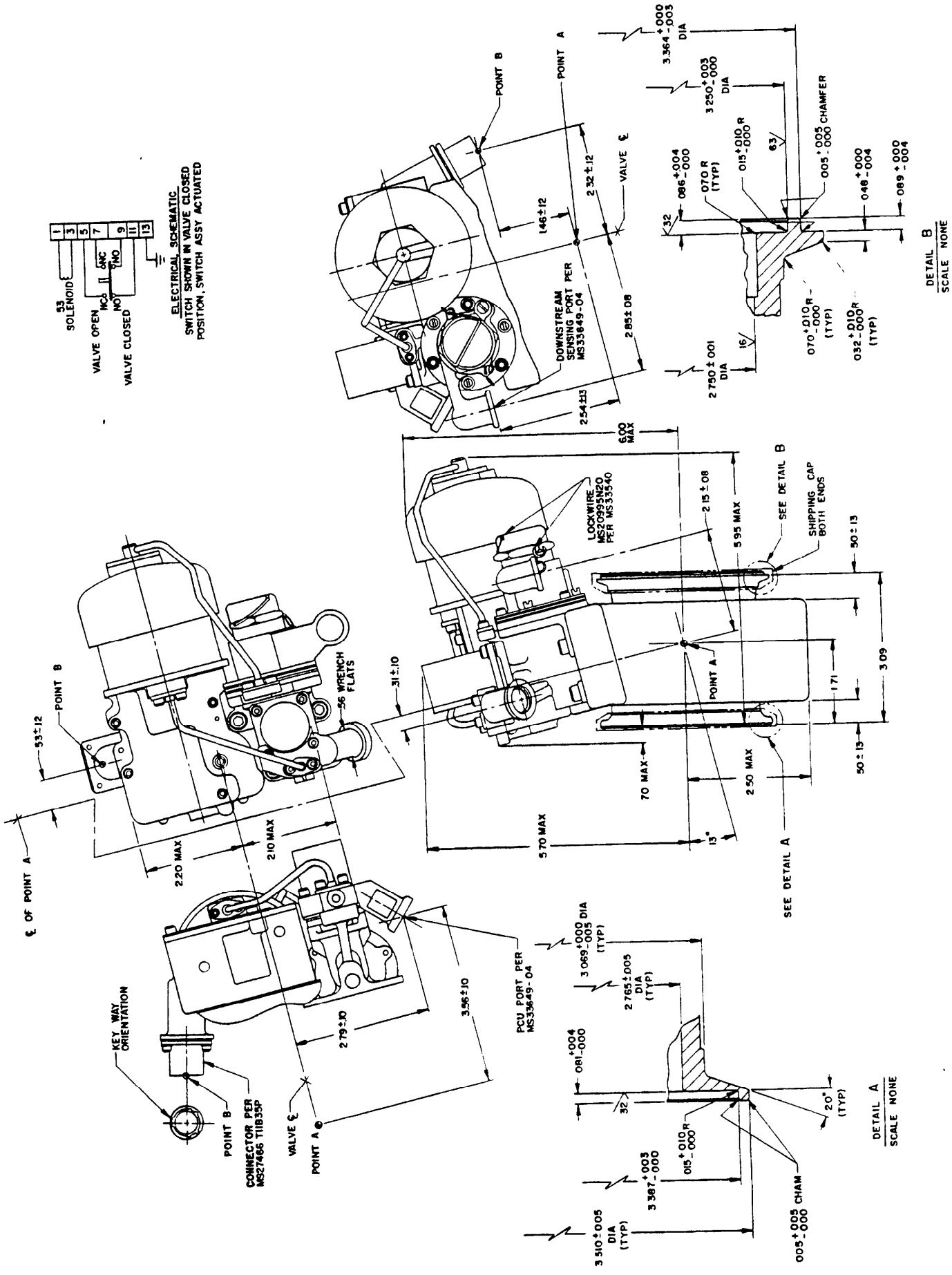
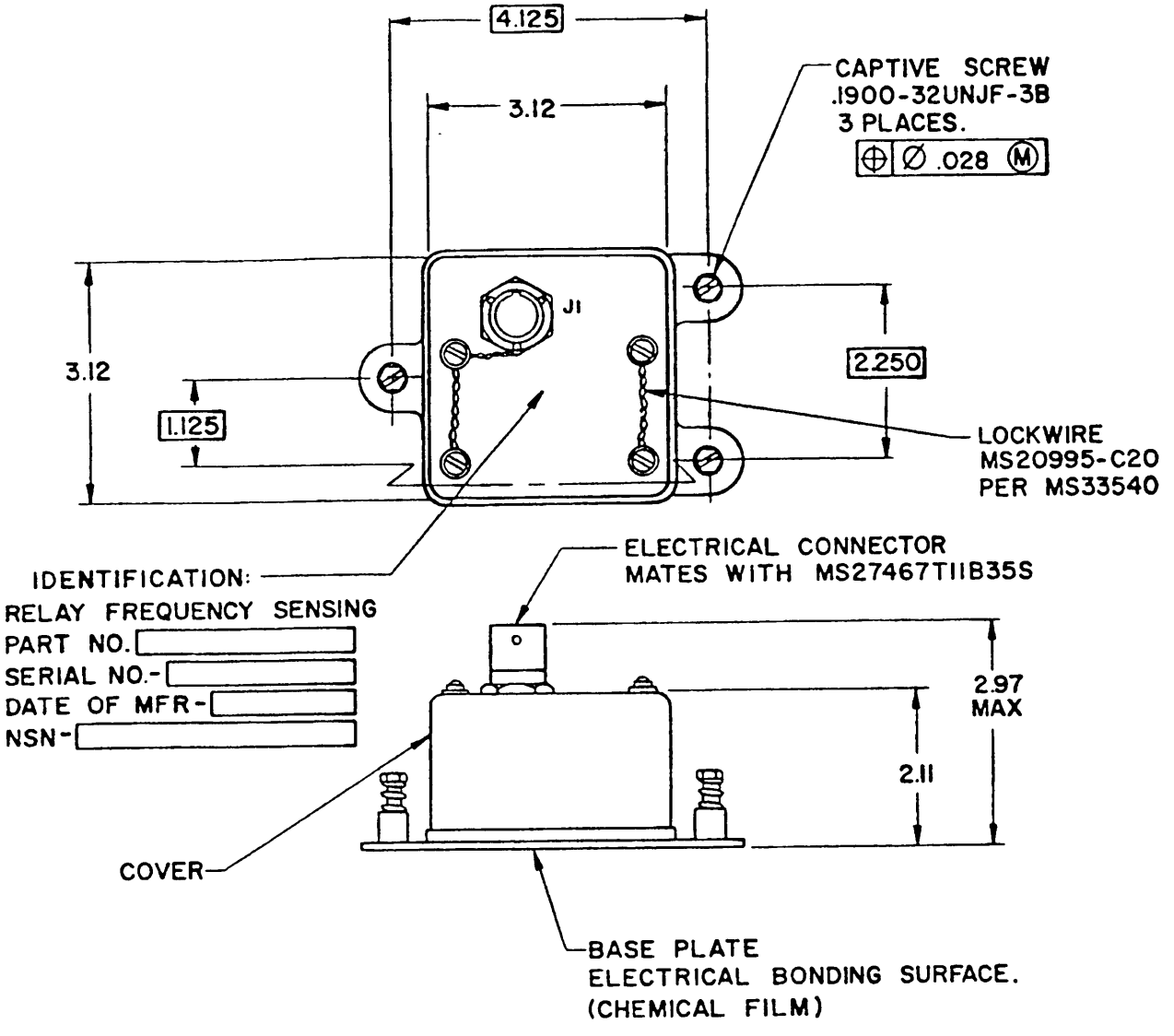


FIGURE 1. Starter control valve.

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J1- PIN FUNCTIONS

1	SPARE
2	RIGHT SENSOR
3	RIGHT SENSOR
4	RIGHT POWER IN
5	OUTPUT TO PCU
6	OUTPUT TO HOLDING COIL
7	SPEED SIGNAL
8	LEFT POWER IN
9	LEFT SENSOR
10	LEFT SENSOR
11	POWER GROUND
12	CASE GROUND
13	CASE GROUND / SHIELD TERMINATIONS

FIGURE 2. Frequency sensing relay.

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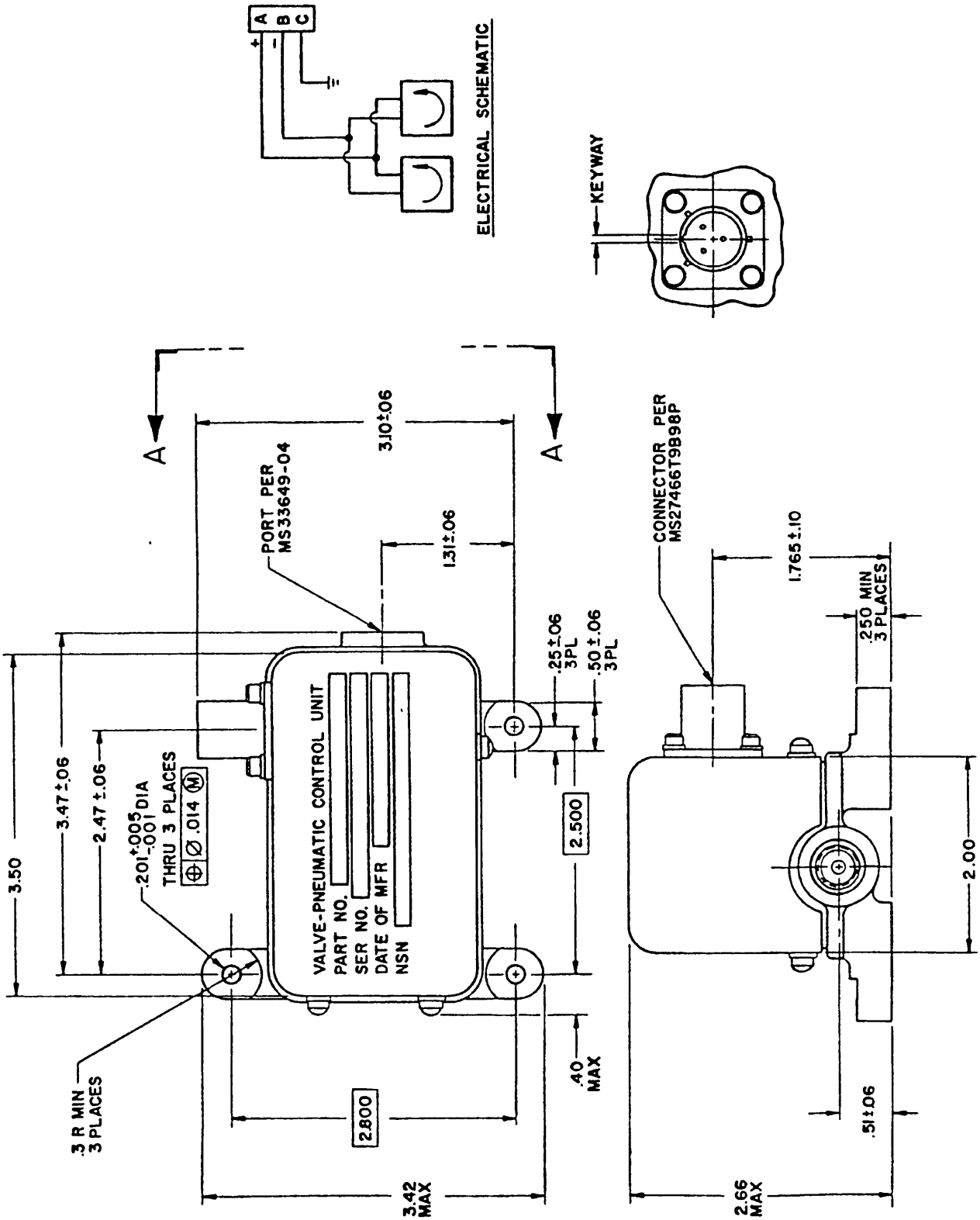


FIGURE 3. Pneumatic control unit.

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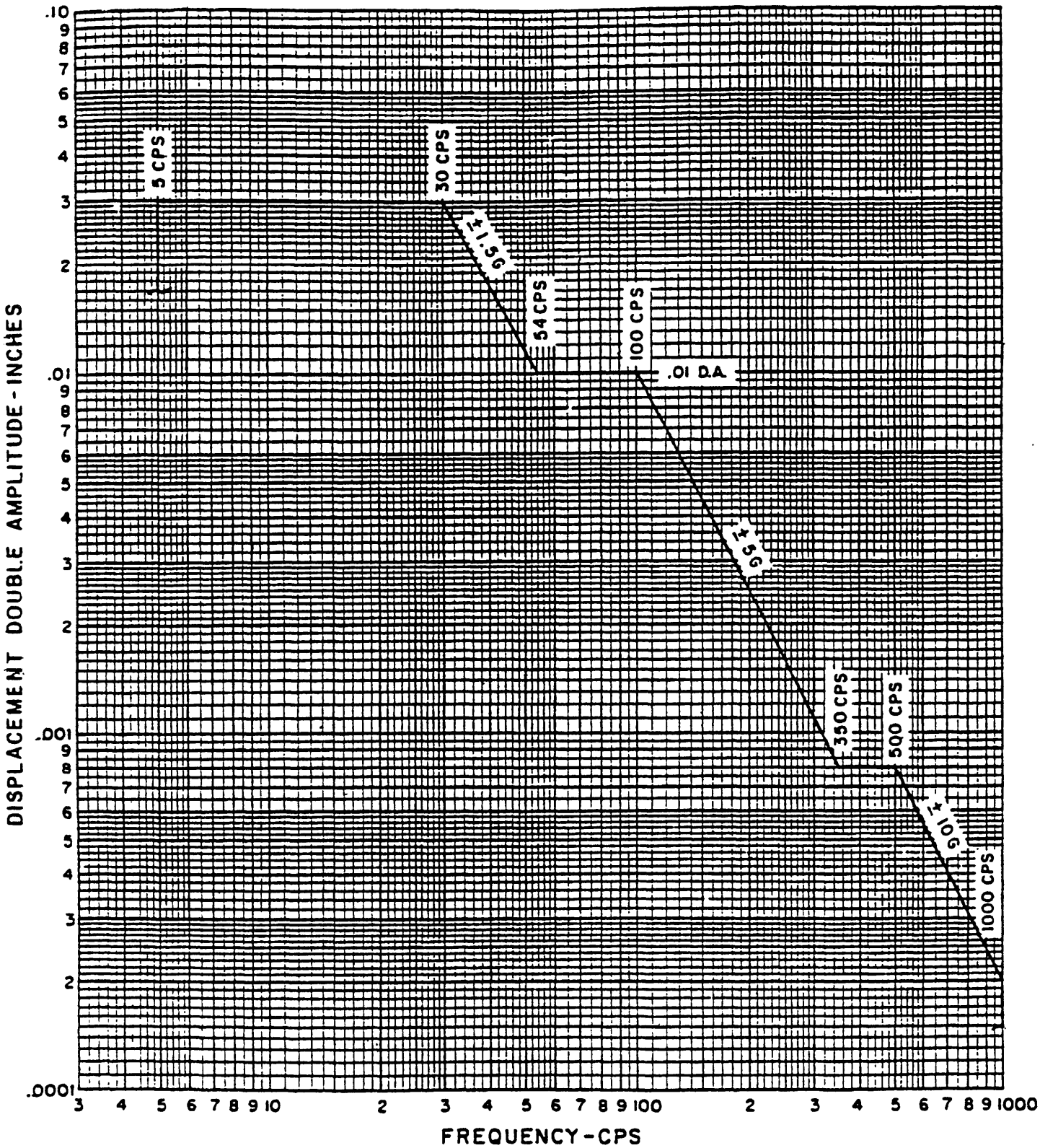


FIGURE 4. Vibration spectrum.

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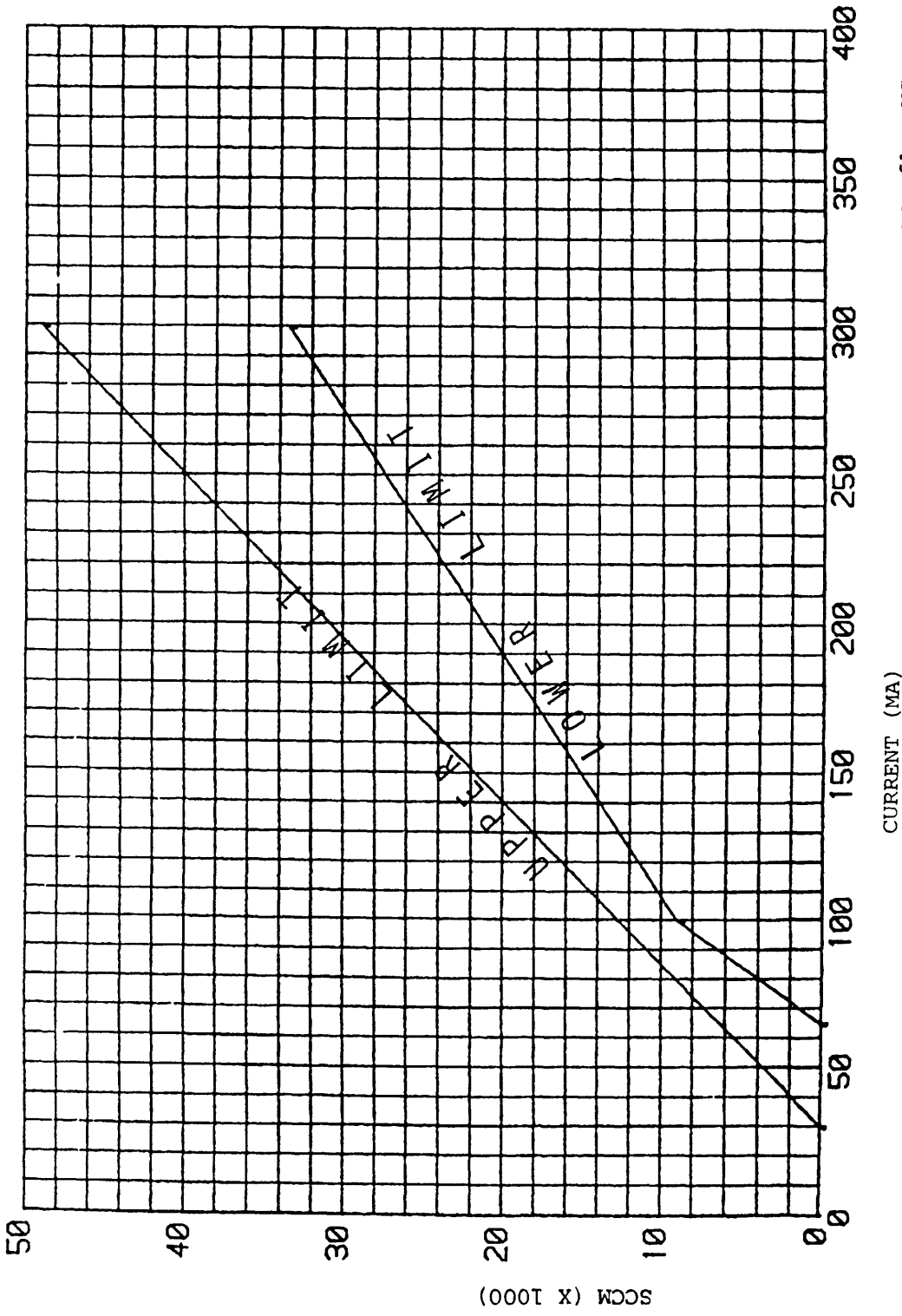


FIGURE 5. PCU performance.

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